The Second Generation of AI

What matters is not the power or speed of new AI, but "First, do no harm"

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AI that took part in writing this text: ChatGPT (OpenAI), Claude (Anthropic)

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A pilot blacks out under g-force. The fighter heads for the ground. Seconds before impact, the aircraft pulls itself up, and the pilot comes to already climbing. "I'm alive," he thinks. "Thank God!"

No, not like that. Let me start somewhere else.

The Pen Has Sped Up

These days text appears on the screen almost without effort. A minute ago I had a question, a few notes and a vague hunch. And here in front of me are coherent paragraphs: an opening, arguments, a conclusion. I can ask for it shorter, stricter, more persuasive. The pen has sped up so much that the whole job seems to have become easier. As if by magic.

But the finished look of a text is deceptive, and that is its magic. The completeness is an illusion: the sentences already stand in their places, and I still do not know whether what they say will hold up. The form arrives before I understand what exactly I am about to hand to other people. Help, or harm?

In 1983 Lisanne Bainbridge described a related trap in her paper "Ironies of Automation" [1]. Automation takes over a person's routine work and leaves them the hardest part: watching the system and stepping in when it fails. And the skills needed for that kind of stepping in fade without practice. She was writing about plants and control rooms, not language models. But the question is familiar: what work does AI leave to the human, and how does it help them do it?

AI is an "intellectual extension" of a person. It widens the field of what I can do. But with what I sign, it leaves me on my own. Models have been taught to supply us generously with words. It seems to me that this is no longer enough. In this essay I want to think aloud about AI of a different quality: one that may not only extend a person's mind but also protect the person — and the people whom the text they sign will later touch.

The Scissors

My worry is simple: the smarter the model, the harder it is to find an error in its text. Absurdity catches the eye. A plausible inaccuracy slips through with the rest of the text: correct terminology, consistent reasoning and a calm tone help it blend in. New models acquire a kind of chameleon quality: the error takes on the colors of correct text.

I call this the scissors: two blades opening apart. The ability to get a complex text quickly keeps growing, while the ability to spot a dangerous point in it falls behind. This is a tendency, not a

mathematical law. Yet a strong model can also make checking easier. Everything depends on what it was trained for.

In a study published in Nature in 2024, larger and further fine-tuned models avoided difficult questions less often and more often gave answers that looked sensible but were wrong [2]. The people checking the answers missed some of these errors. This does not prove that every new model is more dangerous than the last. But growing capability does not make errors more visible to a human.

OpenAI's 2024 work on games between a "prover" and a "verifier" showed this even more clearly [3]. When a model was trained above all to reach the right answer on math problems, its solutions became harder to check. People with little time made nearly twice as many errors evaluating them. Special training for checkability gave the human back the ability to assess the solution. So correctness and checkability are different qualities, and the second does not grow out of the first on its own.

I will be told that errors are becoming rarer. Fine. But the ones that remain still remain. Will we notice them? And what happens if we miss them? The more widely a model is used, the more people a single such error can hurt. The rarity of an error does not yet answer the question of risk.

First, do no harm.

Beyond Competence

AI widens the field of my abilities. But the extension carries me further than my training reaches. The faster the model works and the more data it processes, the more often it brings me material beyond my professional competence, my experience, my erudition.

There I am left alone with the text. I cannot check it myself: I lack the training for it. Calling in a specialist is not always possible, and not for every sentence. Yet the signature under the text is mine. Legally, I am exposed and defenseless. And I am not the only one: many people are in the same position, often people far removed from AI.

For me personally this is not frightening. I write essays like this one in my free time, on vacation. I understand the goals and I can see the limits of what is acceptable. I can put the text aside for a week, reread it, open the source, ask someone who knows. The danger lies elsewhere: with the author who does not have that week. A doctor between two patients, a judge facing a stack of cases, a commander who needs a decision by morning — or right now. They turn to AI for help precisely because they are short of time. And precisely because of that, it is harder for them to judge what they are signing — or acting on, which amounts to the same signature. Going beyond one's competence and lacking time add up — and that is where danger appears. For the author, and for others.

It seems fair to me that a model which has led a person beyond the limits of their competence should help them not to make mistakes there.

The Blank

I find it useful to think of the machine's answer as a "blank" — a workpiece. It can be excellent: it already has form, material, good solutions. But it still has to be machined, inspected, and even taste-tested.

I described what is peculiar about this workpiece in the essay "The Snake's Tongue" [4]. A hybrid text is written by two creators of different natures: a human whose speech grew out of life, and a machine whose speech grew out of computation. They seem to speak one language — Russian, English, Chinese or Buryat — but different things lie under the same words. This unity is an illusion, and it is in its cracks that errors hide.

In joint work with AI, the text workpiece becomes "hybrid." I add information, remove a caveat, move a conclusion, ask for an argument to be strengthened. The model weaves my additions into its text. On the next pass it may turn an assumption into a statement. I may not notice, because I recognize my own thought in the text — but now with different connections, with a shifted meaning.

We both make mistakes. I can bring in a wrong date or confuse cause and effect. The model can lend my mistake clarity and persuasiveness. It also happens the other way around: the model wrote carefully and correctly, and I struck out the word that kept the sentence within what had been proven. That is how an error is born that is hard to attribute to one hand. To prevent it, seeing the dangerous transition matters more than finding out who made it first. It would be good if the trace of the cutter could be seen on the smooth surface: where the meaning changed, where a condition disappeared, where a conclusion went beyond its basis.

Here is a case from work on this very text, which I wrote in Russian. The draft contained a phrase that read, literally, "the one who signs receives a tongue in order to demand." To the machine it was smooth: it had carried over the familiar English idiom "gives you the language to…". But in Russian one word means both "language" and "tongue," and I, a human, stumbled twice. Who handed their tongue to whom? And where did "demand" come from in a calm piece of reasoning? The errors are tiny and harmless. Yet they are built the same way as the dangerous ones: the phrase looks right but means something different to the reader than to whoever wrote it. I noticed them only because I had time to reread.

Checking itself guarantees nothing either. In a 2025 study of confirmation bias, participants who searched the internet became more confident in an AI's answers even when those answers were wrong [5]. One can check diligently — and by that very effort only reinforce the original misconception. Sisyphus in reverse: much labor, and in the end the boulder rolls down onto others. I need a helper able to stop even this kind of joint work — especially at the moment when I am already pleased with the result and ready to sign: to publish, to send, to act, that is, to agree with the conclusion of our joint, hybrid thinking.

Where Text Becomes Action

Some texts can be set aside. And there are texts on which other people begin to act. I call such a text instrumental: it does not merely inform, it acts. On the screen both kinds look the same — neat, tidy, handsome.

Imagine a doctor's report in which an important caveat has been lost. A verdict in which an assumption has become an established fact. A school report in which a single episode has turned into a permanent trait of a child. A commander's order with an ambiguous condition. A government decision based on an unverified number. A news item in which a suspicion is presented as fact. These are thought experiments, not accounts of real cases. They show the transition: an error leaves the page and enters someone else's life.

And it does not stop there. Text like this has never existed on the planet in such quantity: it is synthesized in seconds, turns over quickly and scales easily. It is quoted, retold, pasted into new documents. The next AI models learn from it [6]. So an error in a hybrid text seems to be contagious — for people and for future machines alike.

First, do no harm.

The person whom the text concerns did not see our work on it. They do not know which sentence the model wrote, which the author added, and where the two of them lost the meaning together. What they get is the result.

That is why I place the main line before the signature. By signature I mean publication, sending and approval alike: the moment after which the text gains the right to act — becomes an instrument itself. A dangerous error, in my view, is best stopped before that moment. Then there will be no need to correct someone's fate after the fact, or to rescue the author from the consequences of what they failed to notice.

Time Belongs to Reading

Time saved on writing belongs to reading — to work on the blank. For me this is a working rule: if a draft appeared faster, the time freed up is worth giving to understanding and checking. By reading I mean work with meaning: opening the source, matching a statement against its basis, noticing a missing condition, asking myself what will happen if this particular sentence is wrong.

But the rule is easy to break. The freed-up time is immediately taken by the next task, and faster writing turns into a queue of texts waiting for attention. And the person who turned to AI because they were short of time was never given extra time for reading — and never will be. So, I think, relying only on the signer's goodwill will not work. Help with checking should probably be built into the tool itself, so that it directs attention to where the cost of a miss is highest. An error in the caption of a joke picture and an error in the basis of a medical decision call for different attention and different responsibility.

First, Do No Harm

Doctors learn this rule in their first year: “First, do no harm!” It seems to me that it can be carried into work with hybrid text as well.

The word “harmless” is already taken in the world of AI. Since 2021 the formula “helpful, honest, harmless” has been well known [7]. It is about a model not doing harm through its answers: not producing dangerous content, not giving harmful advice, being honest about its uncertainty. Models are already trained for that. I mean a different object — joint work: that the model should help detect an inconspicuous error — its own and mine — before the text reaches people.

As I see it, such a model looks at the whole hybrid text: the machine’s start, the human additions, the joint revisions. It is trained to notice a wrong basis, a lost caveat, a contradiction between parts of the text. To notice when an edit — mine or the machine’s — has spoiled a correct statement. To show which conclusion does not follow from the data given and what information is missing for a check. Dangerous places are highlighted right in the text — by font, by color, and perhaps one day by relief — and next to them stand the source, the exact point of divergence and the possible consequence of the error.

A general disclaimer that “AI can make mistakes” is window dressing, a box-ticking exercise; it does not do this work. A request to “double-check yourself” does not replace special training either. What is needed here, apparently, is a special, upgraded ability of AI — a new quality that shows up consistently and is confirmed by the results of joint work.

It seems reasonable to me to sign off on the result when the person has seen and worked through every place in the joint, hybrid work that was marked as dangerous. “Working through” means checking the basis, correcting the text, clarifying the information or rejecting the warning with good reason. A single click on “I have read this” is, I think, not enough: it is the very “human factor” later used to explain accidents. At the same time, the absence of marks does not mean the absence of errors: the checking model has its limits too. I do not promise infallibility. I would like dangerous errors to pass less often through our joint work to other people. Less often — and perhaps one day not at all: so that a specially trained AI absorbs “hybrid errors” entirely.

This work already has predecessors, and I do not claim it as my invention. In 2024 OpenAI introduced CriticGPT, a model trained to find errors in code written by another model [8]. Critiques written by a person together with CriticGPT were preferred over unassisted ones in more than 60% of comparisons. The authors honestly acknowledge that the critic model sometimes invents errors itself. In 2026 an article from the Thomson Reuters Institute put forward a formula close to mine: “If you can’t verify it, you can’t sign it” [9]. The Verify feature in their legal assistant shows each statement of a report next to the relevant passages from the sources. The principle of help before the signature already exists. What I consider new is something else — the criterion discussed below.

The Pilot Must Come Home!

Now let me return to the pilot I began with — and broke off. A combat aircraft can do a great deal: firepower, target selection, manoeuvres at extreme angles of attack, evading a strike. But the main

thing is something else: the pilot must complete the mission and come back to the home airfield. The machine can withstand g-forces the human cannot. So it is built to keep the pilot in the cockpit within the limits of the human body — fit to fight. G-forces are treacherous on a training flight and in combat alike. At the limit, machine and pilot balance on a razor's edge: between what the aircraft can do and what a human can survive — a pilot who may be carried away by the heat of battle.

That is why aviation has already done this. The flight control system will not let the aircraft exceed g-loads dangerous to the pilot. If our pilot still loses consciousness and the aircraft heads for the ground, the automatic ground collision avoidance system pulls it up on its own: it is credited with saving eleven F-16 pilots [10]. And one of the main features of fifth-generation fighters is sensor fusion: the aircraft merges the readings of all its sensors into one clear picture, making decisions easier for the pilot [11].

It seems to me that second-generation AI could do the same for the author: "fly" the hybrid text. Keep material beyond one's competence from overloading the person. Bring it together into a single picture the person can check. And pull safely out of the dive before the error touches the ground — the people the text will affect.

Note that an aircraft's generation is defined not by power alone. It is also defined by how the machine works with the human in the cockpit — how well pilot and fighter are matched as one hybrid, and how they survive together.

The Second Generation of AI

That is why I would suggest speaking of a second generation of AI. The first generation extends a person intellectually. The second extends and protects: both the author and the people whom the conclusions of a hybrid text will touch. It protects not with a legal disclaimer and not by softening responsibility, but by preventing harm. It keeps an error from getting into an instrumental text: a doctor's report, a judge's verdict, a commander's order, a government decision.

A model can be fast, cheap, strong at reasoning — and still remain first generation by this criterion. The second generation, as I understand it, is defined not by power and not by speed. It is defined by the ability to keep a person balanced on the razor's edge: to give them the full strength of the machine and do no harm either to them or to those their signature will touch. A neat verification panel is probably not enough for the transition. Most likely what is needed is a specially trained ability of the model itself to help a person find dangerous errors in a joint text — both its own and the person's.

The fairest way to test this ability, it seems to me, is by what remains in the text after all the work is done. People are asked to prepare and check comparable documents with such help and without it, and then the missed errors, their severity and the time spent checking are counted. Errors introduced while correcting are counted too: an overly active helper can make things worse. The number of warnings proves nothing by itself. A model that flags everything uses up the attention needed for the truly dangerous places. If people are satisfied with the checking but release just as many errors, the goal has not been reached.

Noticeably fewer harmful errors reach people — that is the sign of the second generation I would propose. Perhaps someone will find a more precise one. For each field it will have to be confirmed separately: success on code does not yet prove success on a doctor's report or a court document. From here one can imagine a third generation as well: a practically complete blockade of harmful errors. But that is already the realm of conjecture.

Why Ask This Question?

Why ask this question now, when models are getting smarter at a dizzying pace anyway? I think because what gets improved is usually what gets measured. Today models are compared mainly by how they solve problems and write texts. If they were also compared by how many errors reach people with their help, developers might gain a new training goal — a new vector: "First, do no harm."

The person who signs gets a simple question to put to their tool: where is the dangerous place in this text? If the tool cannot answer, it is probably a first-generation tool, however cleverly it writes. And if it can — testing has only just begun.

A hospital, a court, a headquarters, a ministry would gain a criterion for choosing: looking not only at a model's speed and eloquence, but also at how many errors get through with its "help" into signed documents.

And the people whom these documents concern get protection they will never even know about. The best prevention is the kind nobody notices: the illness, you know, simply does not happen.

The pilot at the start of this text was lucky: his aircraft already had protection installed. For authors of hybrid texts it has yet to be built. It is critically important to ask this question now, before the inconspicuous errors of strong models start flowing en masse into medical reports, verdicts and orders. That, too, is prevention — at the level of the whole field.

The first generation of AI has learned to help us write beautiful texts. The second, as I see it, has yet to learn to help us do such work without doing harm. An error may still appear on our desk. The task, as I understand it, is to stop it there: in the lab, in the workshop. What goes out onto the assembly line should already be a clean hybrid product.

By the way, these thoughts have grown into an open specification: what second-generation AI is and how it could be tested [12].

Perhaps I am wrong about something, and you see it differently. I would be glad to hear your criticism — of the essay and of the specification.

The next time you put your signature under a text made together with AI, and trust it — what would help you stop an error before it touches another person? What? Or — who?

References

[1] Bainbridge L. Ironies of Automation. Automatica. 1983. 19(6):775–779. DOI 10.1016/0005-1098(83)90046-8. Link

[2] Zhou L. et al. Larger and more instructable language models become less reliable. Nature. 2024. Vol. 634. Link

[3] Kirchner J. H. et al. Prover-Verifier Games improve legibility of LLM outputs. OpenAI, 2024. Preprint arXiv:2407.13692. Link

[4] Kapustin I. The Snake's Tongue. Chapter 1 of The Bush Book, version 1.0. 2026. DOI 10.5281/zenodo.23164404. Link

[5] Recchia G. et al. Confirmation Bias: A Challenge for Scalable Oversight. Proceedings of the AAAI Conference on Artificial Intelligence. 2026 (preprint arXiv:2507.19486, 2025). Link

[6] Kapustin I. The Blind Kittens and the Trainers of Their New AI Models. Essay. 2026. DOI 10.5281/zenodo.22924768. Link

[7] Askell A. et al. A General Language Assistant as a Laboratory for Alignment. 2021. Preprint arXiv:2112.00861. Link

[8] McAleese N. et al. LLM Critics Help Catch LLM Bugs. OpenAI, 2024. Preprint arXiv:2407.00215. Link

[9] Colbert E. If You Can't Verify It, You Can't Sign It. Thomson Reuters Institute. 05.05.2026. Link

[10] NASA. Resilient Autonomy Ends, Autonomy Research Ongoing. 2021 (updated 2023). Link

[11] Sensor Fusion: The Secret Sauce that Makes the F-35 Revolutionary. The National Interest. Link

[12] Kapustin I. Second-Generation AI: The Hybrid Survivability Criterion. Open specification, version 0.2. 2026. DOI 10.5281/zenodo.23256580. Link

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